12TH BIOLOGY · CHAPTER 22 · FULL CHAPTER MCQS

Biostatistics

55 topic-wise MCQs from the complete chapter with answers and reasons

22.1 BIOSTATISTICS

1. Biostatistics applies statistical methods to:

(a) Financial data
(b) Biological, medical and health-related data
(c) Weather data only
(d) Engineering data

Answer: (b) Biological, medical and health-related data — This is its definition.

2. Biostatistics draws conclusions about a population by using:

(a) The whole population
(b) Only old records
(c) Guesswork
(d) Sample data

Answer: (d) Sample data — Testing a whole population is impractical or costly.

3. Randomization and calculating sample size are parts of:

(a) Data collection
(b) Data management
(c) Study design
(d) Presentation

Answer: (c) Study design — Study design and experimental planning.

4. Medical records and government reports are sources of:

(a) Secondary data
(b) Primary data
(c) Experimental data
(d) Control data

Answer: (a) Secondary data — They are existing data.

5. Clinical trials and interviews give:

(a) Secondary data
(b) Coded data
(c) Primary data
(d) Inferential data

Answer: (c) Primary data — The data are collected directly.

6. Data cleaning, coding and storage belong to:

(a) Data analysis
(b) Interpretation
(c) Study design
(d) Data management and processing

Answer: (d) Data management and processing — Done before analysis.

7. Mean, median and mode are examples of:

(a) Inferential statistics
(b) Descriptive statistics
(c) Regression analysis
(d) Modelling

Answer: (b) Descriptive statistics — They describe the data.

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8. Hypothesis testing and regression analysis are examples of:

(a) Inferential statistics (b) Descriptive statistics
(c) Data collection (d) Data coding

Answer: (a) Inferential statistics — They draw conclusions about populations.

9. The study of distribution and determinants of diseases in populations is:

(a) Genomics (b) Epidemiology
(c) Demography (d) Bioinformatics

Answer: (b) Epidemiology — Biostatistics is used in it.

10. Analysing vast genomic and proteomic datasets is the work of:

(a) Bioinformatics (b) Demography
(c) Epidemiology (d) Quality control

Answer: (a) Bioinformatics — It supports drug discovery and diagnosis.

11. Trends in births and deaths are analysed in:

(a) Genomics (b) Clinical trials
(c) Bioinformatics (d) Demography

Answer: (d) Demography — Health risk assessment and demography.

22.2 KEY STATISTICAL MEASURES

12. The mean is also known as the:

(a) Middle value (b) Most frequent value
(c) Average (d) Spread

Answer: (c) Average — It is a measure of central tendency.

13. The mean of 400, 420, 470, 390 and 460 is:

(a) 428 (b) 420
(c) 430 (d) 2140

Answer: (a) 428 — 2140 ÷ 5 = 428.

14. In the grouped-data formula of mean, x_i is the:

(a) Frequency (b) Midpoint of the class interval
(c) Class width (d) Lower boundary

Answer: (b) Midpoint of the class interval — f_i is the frequency.

15. The midpoint of the class interval 20–30 is:

(a) 20 (b) 50
(c) 30 (d) 25

Answer: (d) 25 — (20 + 30) ÷ 2.

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16. The middle value of an ordered dataset is the:

(a) Median (b) Mean
(c) Mode (d) Range

Answer: (a) Median — It is found after arranging the data in order.

17. The median of 50, 60, 65, 75 and 80 is:

(a) 60 (b) 66
(c) 65 (d) 75

Answer: (c) 65 — The 3rd of five ordered values.

18. The median of 300, 315, 330, 400, 450 and 500 is:

(a) 330 (b) 382.5
(c) 400 (d) 365

Answer: (d) 365 — (330 + 400) ÷ 2.

19. For an even number of values, the median is the:

(a) Larger middle value (b) Average of the two middle values
(c) Smaller middle value (d) Most frequent value

Answer: (b) Average of the two middle values — Positions n/2 and (n/2) + 1.

20. The mode of 50, 50, 50, 50, 55, 55, 60, 60, 65, 70 is:

(a) 55 (b) 70
(c) 60 (d) 50

Answer: (d) 50 — 50 appears four times.

21. In the mode formula for grouped data, h stands for:

(a) Class width (b) Highest frequency
(c) Height (d) Lower boundary

Answer: (a) Class width — Upper limit minus lower limit of a class.

22. The class with the highest frequency is the:

(a) Median class (b) Modal class
(c) Mean class (d) First class

Answer: (b) Modal class — It is used in the mode formula.

23. Standard deviation measures the:

(a) Central value (b) Most frequent value
(c) Dispersion of data about the mean (d) Total of values

Answer: (c) Dispersion of data about the mean — It is a measure of variability.

24. A standard deviation close to zero means the data points are:

(a) Widely spread (b) Not measurable
(c) All negative (d) Very close to the mean

Answer: (d) Very close to the mean — The curve is narrow.

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25. The average of the squared deviations is called the:

(a) Variance (b) Range
(c) Percentile (d) Mode

Answer: (a) Variance — Its square root is the standard deviation.

26. If the variance is 87.11, the standard deviation is about:

(a) 8.71 (b) 43.5
(c) 9.33 (d) 87.11

Answer: (c) 9.33 — √87.11 = 9.33.

27. Deviations are squared in order to:

(a) Make them smaller (b) Eliminate negative values
(c) Find the median (d) Find the range

Answer: (b) Eliminate negative values — It also emphasizes larger deviations.

28. The simplest measure of variability is the:

(a) Standard deviation (b) Variance
(c) Range (d) Percentile

Answer: (c) Range — Maximum minus minimum.

29. The range of 20, 22, 25, 30, 42, 45, 50 is:

(a) 30 (b) 25
(c) 50 (d) 70

Answer: (a) 30 — 50 – 20.

30. The 25th percentile is also called the:

(a) Median (b) Mode
(c) Third quartile (d) First quartile

Answer: (d) First quartile — Q1.

31. The 75th percentile is the:

(a) First quartile (b) Third quartile
(c) Median (d) Range

Answer: (b) Third quartile — Q3.

32. For n = 10, the rank of the 40th percentile is:

(a) 4.0 (b) 0.4
(c) 5.0 (d) 4.4

Answer: (d) 4.4 — R = (40/100) × (10 + 1).

33. When the rank R is not an integer, the percentile is found by:

(a) Rounding off (b) Interpolation
(c) Taking the mode (d) Ignoring it

Answer: (b) Interpolation — Between the two nearest ranks.

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22.3 CHARTS

34. In a bar chart, categories are usually shown on the:

(a) x-axis (b) y-axis
(c) Legend (d) Title

Answer: (a) x-axis — Values are on the y-axis.

35. In a bar chart, the height of a bar is proportional to the:

(a) Category name (b) Number of bars
(c) Width of the bar (d) Value it represents

Answer: (d) Value it represents — So bars can be compared.

36. A bar chart is best for:

(a) Parts of a whole
(b) Comparing quantities across categories
(c) Showing a circle
(d) Finding the median

Answer: (b) Comparing quantities across categories — For example counts of different species.

37. If the greatest value is 120, a suitable y-axis scale is:

(a) 0 to 100 (b) 0 to 12
(c) 0 to 140 (d) 100 to 120

Answer: (c) 0 to 140 — With an increment of 20.

38. The key that explains colours and symbols on a chart is the:

(a) Title (b) Scale
(c) Axis (d) Legend

Answer: (d) Legend — It makes the chart easier to interpret.

39. Each slice of a pie chart is called a:

(a) Bar (b) Sector
(c) Axis (d) Cap

Answer: (b) Sector — Its angle is proportional to its share.

40. In a pie chart of 100 students, 30 students with blood type A get an angle of:

(a) 108° (b) 30°
(c) 72° (d) 144°

Answer: (a) 108° — (30/100) × 360°.

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41. A category forming 40% of the whole has a sector angle of:

(a) 40° (b) 108°
(c) 144° (d) 72°

Answer: (c) 144° — 0.40 × 360°.

42. A pie chart is ideal for showing:

(a) Trends over time (b) Portions of a whole
(c) Error in data (d) Two variables

Answer: (b) Portions of a whole — The whole circle is 100%.

43. The x-axis most commonly represents the:

(a) Dependent variable (b) Legend
(c) Control variable (d) Independent variable

Answer: (d) Independent variable — The dependent variable is on the y-axis.

44. In the enzyme example, the variable on the x-axis is:

(a) Temperature (b) Enzyme activity
(c) Time (d) pH

Answer: (a) Temperature — It is the independent variable.

45. Range error bars extend from the:

(a) Mean to the maximum (b) Zero to the mean
(c) Minimum to the maximum value (d) Median to the mode

Answer: (c) Minimum to the maximum value — They show the full spread.

46. SD error bars extend above and below the:

(a) Minimum (b) Mean
(c) Maximum (d) Origin

Answer: (b) Mean — By one or more standard deviations.

47. The horizontal line at each end of an error bar is the:

(a) Legend (b) Sector
(c) Cap (d) Scale

Answer: (c) Cap — It is added for clarity.

EXPERIMENTAL DESIGN

48. The factor that the experimenter changes is the:

(a) Dependent variable (b) Control group
(c) Control variable (d) Independent variable

Answer: (d) Independent variable — Its effect is observed.

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49. The factor that is measured in an experiment is the:

(a) Independent variable (b) Control variable
(c) Dependent variable (d) Constant

Answer: (c) Dependent variable — It responds to the independent variable.

50. Factors kept the same throughout an experiment are:

(a) Dependent variables (b) Control variables
(c) Independent variables (d) Hypotheses

Answer: (b) Control variables — They ensure a fair test.

51. The group that does not receive the treatment is the:

(a) Control group (b) Experimental group
(c) Dependent group (d) Sample group

Answer: (a) Control group — It is the baseline.

52. In the albino mice experiment, the independent variable is:

(a) Weight gain (b) Age of mice
(c) Type of diet (d) Temperature

Answer: (c) Type of diet — Weight gain is the dependent variable.

53. In the albino mice experiment, the control group receives:

(a) High-fat diet (b) A drug
(c) No food (d) Normal diet

Answer: (d) Normal diet — The experimental group gets the high-fat diet.

54. The first step of experimental design is to:

(a) Draw conclusions (b) State the question
(c) Analyse results (d) Set up groups

Answer: (b) State the question — It defines what you want to find out.

55. A hypothesis is accepted or rejected on the basis of:

(a) The results (b) The question
(c) The title (d) The legend

Answer: (a) The results — The last step of the design.

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